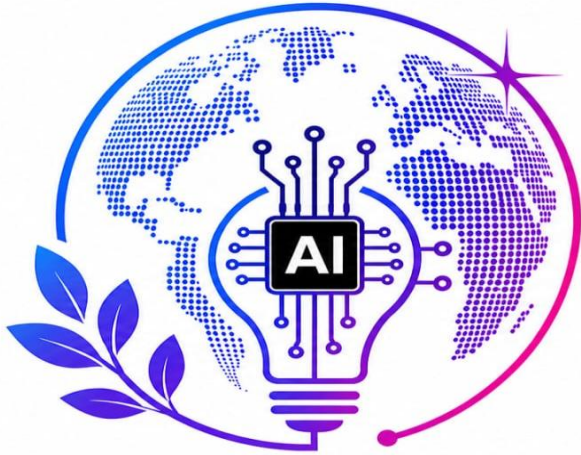




INNOVATE 4 IMPACT

AI SDG GLOBAL
HACKATHON 2026

Innovate today for tomorrow's impact



Team ID: I4I 505

Team Name: Rocket Coders

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University/Institute/College: JSPM's Rajarshi Shahu College of Engineering

Theme: Healthcare, Wellbeing & Accessibility

P.S Category: Software

P.S ID: PS-B03



Team Members

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IDEA TITLE

- Detailed explanation of the proposed solution :- V-TACS is a real-time emergency coordination platform that connects **victim location, symptoms, hospital ASV stock, ventilator availability and ambulance status** to identify and coordinate the most suitable treatment facility. Voice-assisted triage converts spoken emergency information into structured data, while routing provides travel-time estimates
- How it addresses the problem :- Instead of directing victims to the **nearest hospital**, V-TACS identifies a **capable hospital with required resources** and coordinates an available ambulance, helping minimize avoidable delays caused by unavailable antivenom, critical-care support or transport.
- Innovation and uniqueness of the solution :- **Capability-aware emergency coordination combining:**
 -  **Live ASV stock**
 -  **Ventilator availability**
 -  **Ambulance matching**
 -  **Route & ETA calculation**
 -  **AI-assisted voice triage**

TECHNICAL APPROACH

- Technologies to be used (e.g. programming languages, frameworks, hardware) -

Frontend: React + Vite

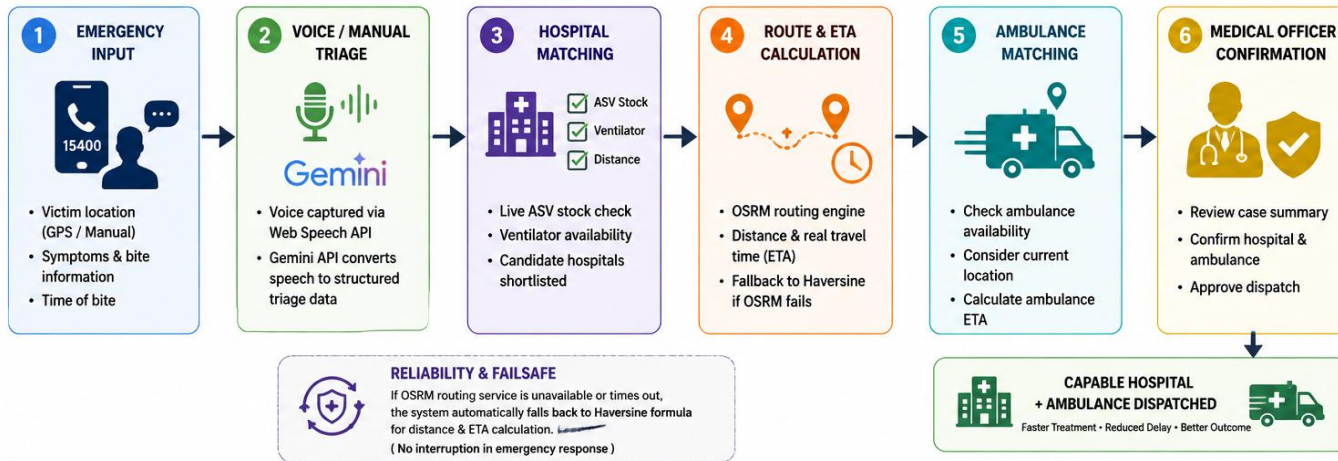
Backend: Node.js + Express.js

Database: MySQL (Aiven)

AI & Voice: Gemini API + Web Speech API

Routing: OSRM + OpenStreetMap

- Methodology and process for implementation (Flow Charts/Images/ working prototype) –





FEASIBILITY AND VIABILITY



- Analysis of the feasibility of the idea :-
 - **Technically feasible:** Uses proven technologies — React, Node.js, MySQL, OSRM and Web Speech/Gemini APIs.
 - **Scalable architecture:** Separate frontend, backend and database layers support future expansion.
 - **Reliable routing:** OSRM provides route/ETA calculation with a **Haversine fallback** if the routing service fails.
 - **Prototype-ready:** The system can be demonstrated using seeded hospital and ambulance data centered around Pune.
- Potential challenges and risks :-
 - **Real-time ASV data availability** from hospitals.
 - **Inaccurate/outdated ambulance or hospital status.**
 - **Internet/API failure** during an emergency.
 - **AI transcription errors** during voice triage.
 - **Security & privacy risks** involving emergency/medical data.
 - **Adoption challenge:** Hospitals and emergency services need to actively maintain their data
- Strategies for overcoming these challenges :-
 - **Live data validation & regular stock updates** for hospital resources.
 - **Fallback routing:** OSRM → Haversine calculation during API failure.
 - **Human-in-the-loop:** Medical officer must explicitly confirm dispatch/resource actions; AI cannot directly execute them.
 - **Secure architecture:** JWT authentication, bcrypt hashing, server-side API keys, rate limiting and input validation.
 - **Graceful degradation & monitoring:** Logging of critical ASV reservation and ambulance-dispatch events.



IMPACT AND BENEFITS

- Potential impact on the target audience :-

Target Audience	Impact
 Snakebite Victims	Faster access to a capable hospital
 Medical Officers	Real-time view of ASV, ventilators & cases
 Hospitals	Better resource & case coordination
 Ambulance Services	Faster ambulance matching & ETA
 Rural Communities	Improved access to timely treatment

- Benefits of the solution (social, economic, environmental, etc.) :-

Social	Economic	System / Environmental
Faster access to appropriate care	Reduces avoidable transfers and resource wastage	Reduces unnecessary ambulance/hospital movement
Supports rural & semi-urban communities	Better utilization of available ASV and ambulances	Digital coordination reduces dependence on manual communication
Improves emergency coordination	Helps prevent resources being sent to unsuitable facilities	Creates a foundation for a connected emergency network
Human-controlled AI improves trust	Better visibility of scarce medical resources	Can scale across hospitals and regions



RESEARCH AND REFERENCES

- Details / Links of the reference and research work:-

REFERENCES & RESEARCH

- **National Action Plan for Prevention and Control of Snakebite Envenoming (NAPSE)**
Policy foundation for snakebite prevention, treatment and emergency coordination.
- **OpenStreetMap + OSRM**
Used as the basis for route and travel-time calculation.
- **Google Gemini API Documentation**
Reference for AI-assisted voice-to-structured-data processing.
- **Web Speech API – MDN Web Docs**
Reference for browser-based speech recognition and voice interaction.
- **V-TACS Project Blueprint**
Primary reference for system architecture, database, security, routing, AI workflow and implementation.

Links

- NAPSE: ***mohfw.gov.in***
- OpenStreetMap: ***openstreetmap.org***
- OSRM: ***project-osrm.org***
- Gemini API: ***ai.google.dev***
- Web Speech API: ***developer.mozilla.org***